



Thermodynamics and activity coefficients at infinite dilution for organic solutes in the ionic liquid 1-butyl-1-methylpyrrolidinium dicyanamide

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ABSTRACT

The activity coefficients at infinite dilution, γ_{13}^{∞} , for 46 solutes, including alkanes, cycloalkanes, alkenes, alkynes, aromatic hydrocarbons, alcohols, thiophene, ethers, ketones, esters, aldehyde, acetonitrile, pyridine and 1-nitropropane in the ionic liquid (IL) 1-butyl-1-methylpyrrolidinium dicyanamide, [BMPYR][DCA] were determined by gas-liquid chromatography at five temperatures in range of (313.15–373.15) K at pressure, $p = 101$ kPa. The thermodynamic functions at infinite dilution as partial molar excess Gibbs free energy, $\Delta G_1^{E,\infty}$, enthalpy $\Delta H_1^{E,\infty}$, and entropy term $T_{ref} \Delta S_1^{E,\infty}$ and the gas-liquid partition coefficients, K_1 at infinite dilution were calculated from the experimental γ_{13}^{∞} values obtained over the temperature range. The values of selectivity and capacity for few separation problems as *n*-heptane/benzene, *n*-heptane/thiophene, *n*-heptane/1-heptyne, and ethylbenzene/styrene at $T = 323.15$ K were calculated from γ_{13}^{∞} and compared to literature values for [DCA]-based ionic liquids and [BMPYR]-based ILs. In comparison with the former measured [BMPYR][TCM], or [BMPYR][SCN] ILs, the [DCA]-based IL presents high selectivities for the separation of aromatic hydrocarbons from aliphatic hydrocarbons, and especially for the extraction of thiophene, or pyridine from *n*-heptane with slightly lower capacity than those for [BMPYR][TCM], or [BMPYR][SCN]. New data shows that [BMPYR][DCA] IL may be proposed as an alternative solvent for the separation of aliphatic from aromatic hydrocarbons.

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1. Introduction

Ionic liquids (ILs) with their highly desirable properties, which have attracted the attention researchers and industrial processes, include their low volatility, thermal stability, specific solvation properties and wide electrochemical windows are used in many applications [1–5]. ILs are highly tunable and may be designed for the specific task and separation problems [6]. Pyrrolidinium-based ILs were found to have practical value in many applications because of their easy synthesis procedures, high selectivities in aliphatic/aromatic separation processes and possibilities of extraction of sulfur compounds from fuels [7–20]. The desulfurization process of fuels is based on the extraction of the sulfur compounds, such as

thiophene, benzothiophene, dibenzothiophene, dimethyl disulfide, which are more soluble in the IL than in an aliphatic hydrocarbons [3,21,22]. However, the new idea is to use the method of oxidation and desulfurization [23].

On the other side, it was found that ILs with cyano group, CN[−] in the molecule, and especially in an anion has huge influence on the selectivity of separation of aliphatics/aromatics, olefins/paraffins sulfur compounds/alkanes and many others separations [24–30]. The dicyanamide anion, [DCA][−] has been studied recently by our group in the hex-1-ene/hexane, cyclohexene/cyclohexane and benzothiophene/styrene separations showing large selectivities and acceptable capacities or solute distribution ratios [24–27]. The literature review for hex-1-ene/hexane separation, presented by our group in 2015 [28] has confirmed the large selectivity of the [DCA][−]-based ILs. The selectivity higher than two was observed in limiting activity coefficients measurements, γ_{13}^{∞} for 1-ethyl-3-methylimidazolium dicyanamide, [EMIM][DCA] [28,29] and for 1-

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